

ALEX DAHL

Rochester, MN

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RESEARCH INTERESTS

Computer Animation, Computer Graphics, Real-time Simulation, Soft Body Simulation

EDUCATION

Doctor of Philosophy in Computer Science, University of Maryland, Baltimore County 2026

Dissertation Title: *Rich Real-Time Soft Body Animation*

Bachelor of Science in Computer Science, University of Minnesota, Twin Cities 2012 - 2016

PAPERS

- [1] Alex Dahl and Adam Bargteil. “Early Termination of Conjugate Gradients for Corotated Finite Elements”. In: *Motion, Interaction and Games*. MIG ’19. Newcastle upon Tyne, United Kingdom: Association for Computing Machinery, 2019. ISBN: 9781450369947. DOI: [10.1145/3359566.3360080](https://doi.org/10.1145/3359566.3360080). URL: <https://doi.org/10.1145/3359566.3360080>.
- [2] Alex Dahl and Adam Bargteil. “Global Momentum Preservation for Position-Based Dynamics”. In: *Motion, Interaction and Games*. MIG ’19. Newcastle upon Tyne, United Kingdom: Association for Computing Machinery, 2019. ISBN: 9781450369947. DOI: [10.1145/3359566.3360078](https://doi.org/10.1145/3359566.3360078). URL: <https://doi.org/10.1145/3359566.3360078>.
- [3] Alex Dahl and Adam Bargteil. “Enhancing Visual Appeal of Rigid Body Animation with Layered Dynamic Deformations”. In Preparation. 2026. MIG ’26.

Posters

- [1] Alex Dahl and Adam Bargteil. “Global Momentum Preservation for Position-based Dynamics”. July 2019. Symposium on Computer Animation 2019.

APPOINTMENTS

Engineering Intern June 2022 - Sept 2022
Internship *Apple*

- Worked on computer graphics problems in character animation/machine learning

Teaching Assistant 2016 - 2023 (intermittent)
Introduction to Computer Graphics (CMSC435/634) *University of Maryland, Baltimore County*

- Held office hours and graded assignments for combined grad/undergrad class.
- Graded assignments for a class of 30-40 students.
- Taught concurrent lectures when professor was unavailable.

Undergraduate Teaching Assistant 2015 - Jan 2016
Intro Computer Science Courses *University of Minnesota, Twin Cities*

- Held office hours and graded assignments.
- Ran programming labs.
- Ran discussion section when instructor unavailable.
- Intro to C/C++ (CSCI1113), Intro to Algs. and Program Dev. (CSCI1913), and Linear Algebra (CSCI2033)

Usability Lab Assistant

Office of Information Technology

2013 - 2016

University of Minnesota, Twin Cities

- Assisted in running user experience testing for a variety of types of software
- Ran recording and tracking equipment in lab
- Recruited and scheduled for concurrent projects
- Developed Excel plugins to automate scheduling work

PROFESSIONAL ACTIVITIES

- DEI and Accessibility Subcommittee Member, ACM SIGGRAPH Research Career Development Committee, 2020–2021
- Team Member, GOFIRST Student Group IGVC Software Team, 2012–2014
- Team Member, GOFIRST Student Group Autonomous Snowplow Software Team, 2014–2017
- Volunteer. Referee and Robot Inspector, FIRST FRC and FTC Competitions, 2012–2017